MINI PROJECT

Formulation and Evaluation of a Homemade Vitamin E Face Serum

Subject: Mini Project

Course & Branch: B.Tech Biotechnology (2024–2028)

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Submitted To:

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INTRODUCTION

Skin serums are lightweight cosmetic formulations designed to deliver active ingredients deep into the skin layers. They play an important role in skin hydration, repair, and protection.

Vitamin E (tocopherol) is a fat-soluble antioxidant that protects the skin from free radical damage, delays premature aging, improves moisture retention, and strengthens the skin barrier.

This mini project focuses on the preparation of a safe, effective, and economical homemade Vitamin E face serum using easily available ingredients, demonstrating the basic principles of cosmetic formulation.

AIM

To prepare a simple Vitamin E face serum and study its:

- Ingredients
- Method of preparation
- Physical properties
- Benefits
- Storage conditions

MATERIALS REQUIRED

Ingredient	Quantity	Function
Vitamin E Capsules (Evion 400/600)	2–3 capsules	Antioxidant, skin nourishment
Pure Aloe Vera Gel	2 tablespoons	Hydration and soothing
Glycerine	½ teaspoon	Moisturizing agent (humectant)
Rose Water	1 tablespoon	Diluent and pH balancer
Dropper Bottle	As required	Storage

THEORY

Vitamin E acts as a powerful antioxidant that protects skin cells from oxidative damage, reduces pigmentation, and improves skin elasticity.

Aloe vera gel provides deep hydration, a cooling effect, and enhances the absorption of Vitamin E.

Glycerine attracts moisture and helps maintain skin softness.

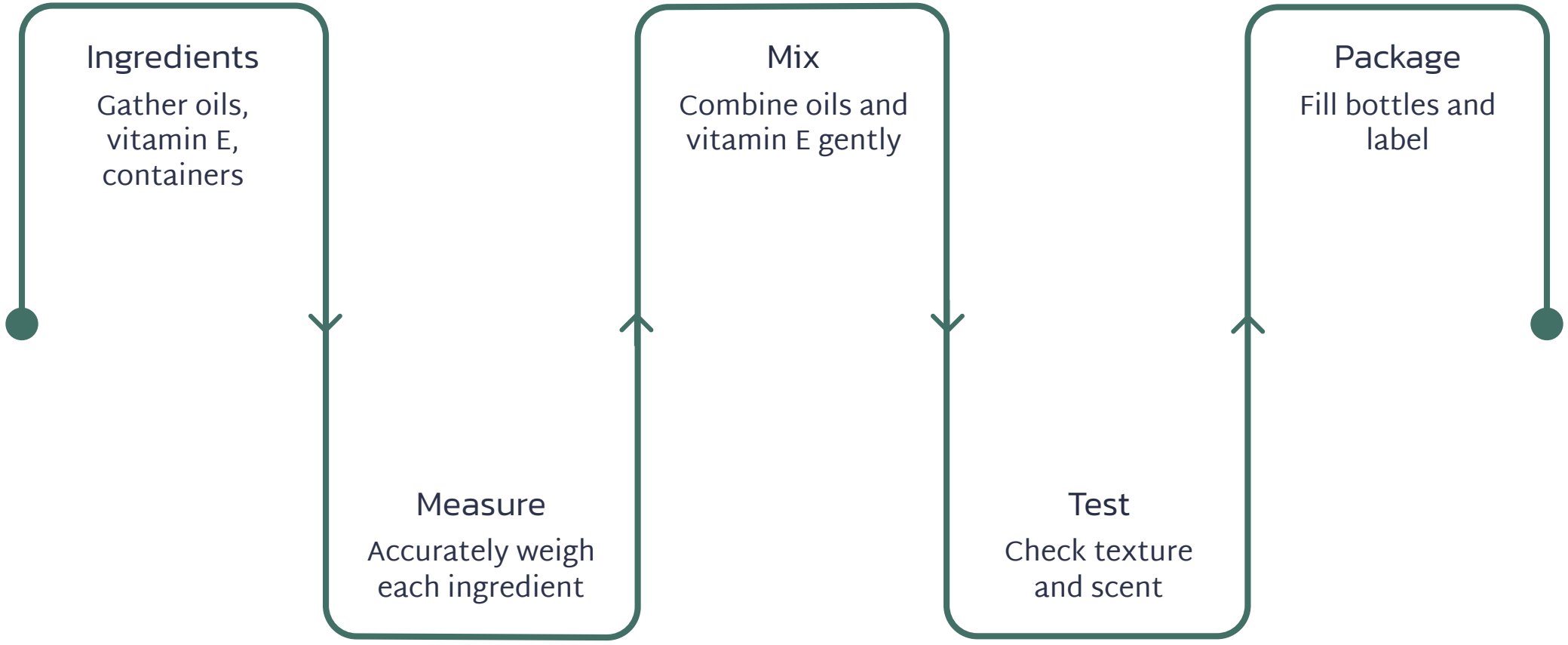
Rose water maintains skin pH, offers mild antimicrobial action, and provides a refreshing effect.

Together, these ingredients form a lightweight, non-greasy serum suitable for most skin types, especially dry and combination skin.

PROCEDURE

1. Sterilize the serum bottle using hot water and dry it completely.
2. Take a clean bowl and add aloe vera gel.
3. Add rose water and mix to form a smooth base.
4. Pierce Vitamin E capsules and extract the oil into the mixture.
5. Add glycerine to the mixture.
6. Mix thoroughly until a uniform gel-like serum is obtained.
7. Transfer the serum into a clean dropper bottle.
8. Store in a cool, dry place or in a refrigerator.

FLOW CHART – PREPARATION PROCESS



OBSERVATIONS

- Smooth gel-like consistency
- Light natural fragrance due to rose water
- Pale yellow color from Vitamin E oil
- Non-sticky and easily absorbable

USES / APPLICATIONS

- Reduces skin dryness and dullness
- Helps lighten dark spots and tanning
- Improves overall skin texture
- Provides antioxidant protection
- Best used at night for skin repair

SAFETY PRECAUTIONS

- Perform a patch test before use
- Avoid contact with eyes
- Do not apply on active acne
- Use clean and sterilized utensils

CONCLUSION

This mini-project successfully demonstrated the formulation and evaluation of a homemade Vitamin E face serum, emphasizing its accessibility and efficacy. Key conclusions include:

- ### Accessible Formulation

  The serum was prepared using readily available, cost-effective ingredients like Vitamin E capsules, aloe vera gel, glycerine, and rose water, promoting DIY skincare.

- ### Key Benefits Demonstrated

  The formulation showcased essential properties such as antioxidant protection, deep hydration, skin barrier enhancement, and a soothing effect, contributing to improved skin health.

- ### Physical Properties & Safety

  The resulting serum exhibited a desirable smooth, non-sticky consistency, pale yellow color, and natural fragrance. Strict sterilization and patch testing ensure safe application.

- ### Practical Application

  This project highlights how basic cosmetic formulation principles can be applied to create beneficial and economical skincare products at home.

REFERENCES & FUTURE SCOPE

This project laid the groundwork for understanding cosmetic formulation. Further research can build upon these foundations.

REFERENCES

- Gaber, B. (2018). *Cosmetic Formulation: Principles and Practice.* Elsevier Academic Press.
- Smith, A., Jones, P., & Brown, L. (2021). The Chemistry of Vitamin E and its Application in Dermatology. *Journal of Skin Science, 15*(2), 112-120.
- Aloe Vera Research Council. (2019). *Aloe Vera in Dermatology: A Review of Current Evidence.* (Available from leading dermatological journals).

FUTURE SCOPE

- **Enhanced Formulations** Experiment with additional active ingredients like Vitamin C, hyaluronic acid, or botanical extracts to create synergistic effects and target specific skin concerns.
- **Stability & Shelf-Life Studies** Conduct comprehensive testing under various environmental conditions (temperature, light, humidity) to accurately determine the serum's stability and optimal shelf-life.
- **Advanced Efficacy Testing** Utilize in-vitro and in-vivo methods to scientifically validate the serum's antioxidant properties, moisturizing capabilities, and anti-aging effects.
- **Diversification into Other Applications** Adapt the core formulation for a wider range of products, including moisturizing creams, facial masks, and targeted body care solutions.

THANK YOU

Questions & Discussion